[INCH-POUND]
A-A-59174
1 April 1998
SUPERSEDING
(See 7.6)

COMMERCIAL ITEM DESCRIPTION

PRESS, ARBOR, MECHANICAL, HAND OPERATED

The General Services Administration has authorized the use of this commercial item description in preference to A-A-51196, A-A-51197, A-A-51198 and A-A-51199 for all federal agencies.

1. SCOPE. This commercial item description (CID) covers bench and floor mounted, hand operated, mechanical arbor presses which are used to force arbors into and out of holes or to assemble and disassemble press fit parts.

2. CLASSIFICATION. Unless otherwise specified (see 7.3), arbor presses shall be of the following types, sizes, and classes.

2.1 Type. Type refers to the type leverage required. The types described herein are as follows:

Type I - Single Leverage
Type II - Ratchet Leverage
Type III - Compound Leverage
Type IV - Ratchet Leverage, Elevating Knee
Type V - Compound Leverage, Elevating Knee

2.2 Size. The size refers to the pressure capacity of the press. The sizes described herein are as follows:

Size A – 0.5 ton	Size D – 1.5 ton	Size G - 4 ton	Size J - 7 ton	Size M - 12 ton
Size B – 0.75 ton	Size E – 2 ton	Size H - 5 ton	Size K - 8 ton	Size N - 15 ton
Size C – 1 ton	Size F – 3 ton	Size I - 6 ton	Size L - 10 ton	Size O - 25 ton

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, Richmond, VA 23297-5610.

AMSC N/A FSC 3444

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TABLE 1. Arbor press types, sizes and classes.

SIZE	PRESSING CAPACITY (TONS)	CLASS	DIAMETER OF WORK ±1.0 (INCHES)	VERTICAL CAPACITY ±1.0 (INCHES)	TYPE				
					1 SINGLE LEVERAGE	2 RATCHET LEVERAGE	3 COMPOUND LEVERAGE	4 RATCHET W/ELEV. KNEE	5 COMPOUND W/ELEV. KNEE
A	0.5	1	6.50	4.50	Y	-	-	-	-
B	0.75	2	8.00	4.50	Y	-	-	-	-
C	1	3	7.00	4.50	Y	-	-	-	-
		4	7.00	5.00	Y	-	-	-	-
D	1.5	5	9.00	7.75	Y	-	-	-	-
E	2	6	10.00	8.50	Y	-	-	-	-
		7	13.00	8.50	Y	-	-	-	-
F	3	8	11.00	21.80	-	Y	-	-	-
		9	12.00	12.00	Y	Y	-	-	-
		10	12.00	18.00	Y	Y	-	-	-
		11	14.00	11.50	Y	Y	-	-	-
		12	14.00	18.25	-	Y	-	-	-
		13	14.00	22.75	Y	Y	-	-	-
		14	15.00	11.00	Y	-	-	-	-
		15	15.00	12.50	Y	-	-	-	-
		16	15.00	18.00	Y	-	-	-	-
		17	24.00	12.00	Y	Y	-	-	-
		18	29.00	12.00	-	Y	Y	-	-
		19	34.00	12.00	Y	Y	-	-	-
		20	35.00	12.00	-	Y	-	-	-
		21	52.00	16.00	Y	Y	-	-	-
G	4	22	15.00	12.00	-	Y	-	-	-
		23	15.00	18.00	-	Y	-	-	-
		24	15.00	23.00	-	Y	-	-	-
		25	18.00	17.50	Y	Y	-	-	-
H	5	26	18.00	17.50	-	Y	-	-	-
		27	20.00	16.00	-	Y	-	-	-
		28	20.00	26.00	-	-	-	Y	-
		29	21.00	21.75	-	Y	-	-	-
		30	22.00	21.00	Y	Y	-	-	-
I	6	31	15.50	23.00	-	Y	Y	-	-
		32	15.50	15.00	-	Y	Y	-	-
		33	15.50	21.50	-	Y	Y	-	-
		34	21.50	30.00	-	Y	-	-	-
		35	22.00	21.00	-	Y	-	-	-
J	7	36	18.00	31.00	-	-	-	-	-
		37	27.00	32.00	-	Y	-	-	-
K	8	38	15.00	12.00	Y	-	-	-	-
		39	18.00	17.50	-	Y	Y	-	-
L	10	40	25.00	12.00	-	Y	Y	-	-
M	12	41	20.00	19.00	-	Y	Y	-	-
		42	20.00	26.00	-	Y	-	Y	Y
		43	21.50	30.00	-	Y	Y	Y	Y
		44	22.00	21.00	-	Y	Y	-	-
N	15	45	17.00	32.00	-	Y	Y	-	-
		46	24.50	46.00	-	Y	Y	-	-
O	25	47	37.00	47.00	-	Y	Y	-	-

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NOTE: "Y" indicates available commercial units.

2.3 Class. Class refers to a specific combination of pressing capacity, diameter of work, and vertical capacity listed in table 1.

3. SALIENT CHARACTERISTICS

3.1 Frame. The frame shall be of the "C" (gap) type and shall support the ram and ram operating mechanism. The frame table shall have a slot aligned with the axis of the ram which will allow passage of the arbor. The frame shall be designed for both bench and pedestal mounting. When specified (see 7.3), a pedestal and mounting hardware shall be supplied.

3.2 Bolster plate. A circular bolster plate shall be provided with not less than four equally spaced slots of varying widths for the passage of an arbor. The bolster plate shall be removable and shall allow the desired plate slot to be positioned below the ram.

3.3 Ram and ram operating mechanism. The ram shall be of the rack-and-pinion type, and shall have an operating mechanism between the lever and pinion of the type specified. The ram shall be perpendicular to the table surface to an accuracy of 0.025 inch total indicator reading (TIR) per 6 inches of ram travel. The leverage ratio shall be of the manufacturer's standard ratio offered to the commercial market for the type specified.

3.4 Leverage mechanism. The leverage mechanism shall be of the type specified. The leverage mechanism shall conform to the requirements of 3.4.1 to 3.4.3, depending on the type of leverage specified.

3.4.1 Single leverage. Single leverage presses shall have an adjustable handle to allow for convenient handle positioning.

3.4.2 Ratchet leverage. Ratchet leverage presses shall have a ratchet and pawl between the lever and pinion. A counter weight shall be mounted to the lever and ratchet for quick return of the ram after pressing is complete. A hand wheel shall be provided for moving the ram up or down while positioning workpieces.

3.4.3 Compound leverage. Compound leverage presses shall be provided with a means for engaging or disengaging the leverage mechanism for single or compound leverage. When specified (see 7.3), a hand wheel shall be provided for moving the ram up or down while positioning workpieces.

3.5 Elevating knee. Elevating knees shall be adjustable for the desired working heights. The method for raising and lowering the knee shall be the manufacturer's standard design offered to the commercial market. The press shall have a device to catch the falling arbor as it is pressed out of the workpiece. The device shall be of adequate size to accommodate 4-1/2 inch arbors and shall be securely mounted on the machine base in alignment with the ram axis.

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3.6 Work area. Work area is defined as the diameter of work times vertical capacity of work. Unless otherwise specified (see 7.3), the size, and work area for the presses to be furnished shall conform to table 1. Tolerances of the horizontal and vertical axis shall be ± 1.0 inch.

3.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. During the manufacture of the press, materials hazardous to the ecological systems, as prohibited by the Federal, State, or local statutes in effect on the date of the contract, shall not be used or emitted.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. PACKAGING.

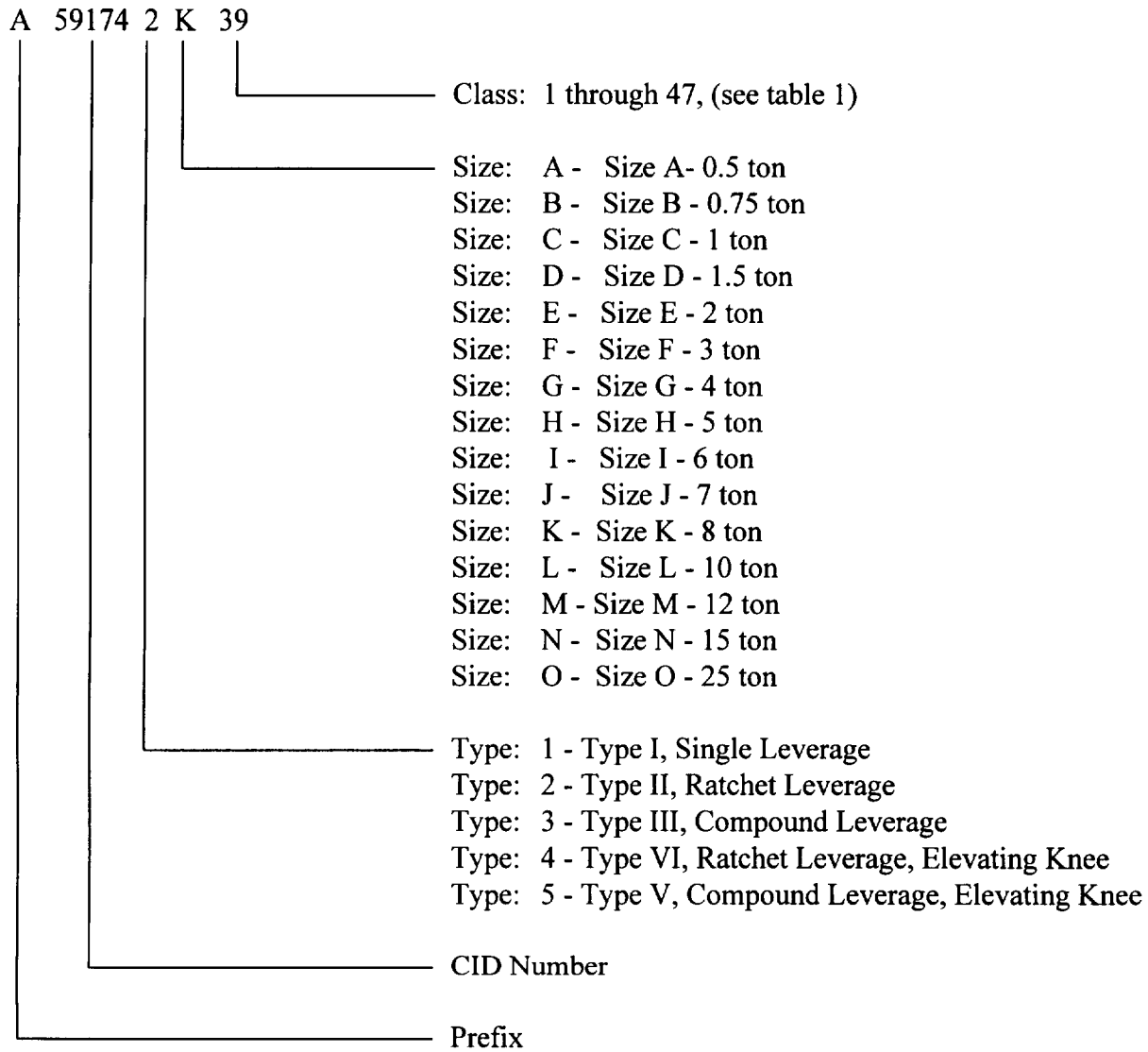
Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59174. Refer to table 1 for type, size, and class designations.

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7.2 Referenced documents.

7.2.1 FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.2.2 The Federal Acquisition Regulation (FAR) maybe obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.3 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in the contract or order:

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a. Title, number, and date of this CID.
b. Type (see 2.1).
c. Size (see 2.2).
d. Class (see 2.3).
e. Quantity required.
f. Packaging, if different (see 6).
g. Pedestal, if required (see 3.1).
h. Hand wheel, if required (see 3.4.3).
i. Size and work area, if different (see 3.6).

7.4 National stock numbers (NSNs). The following NSNs correspond to this CID. The list is not indicative of all possible NSNs associated with this document.

NSN	Old PIN	New PIN
3444-00-223-8353	A51199345	A591741F15
3444-00-223-8357	A51198813	A591743K39
3444-00-243-2652	A51197513	A591742H26
3444-00-254-2114	A5119911222	A591741D5
3444-00-262-4242	A511981017	A591743L40

7.5 Intended use. Presses covered by this CID are for use in machine production, repair, and maintenance shops.

7.6 Supersession data. This CID supersedes A-A-51198, dated February 19, 1991; A-A-51199, dated February 19, 1991; A-A-51196, dated February 20, 1991; and A-A-51197, dated February 20, 1991.

MILITARY INTERESTS:

Custodians
Army - AL
Navy - SH
Air Force - 99

Reviewers
Navy - YD
Air Force - 84

CIVIL AGENCY COORDINATION ACTIVITY
GSA - FSS

PREPARING ACTIVITY
DLA - GS

Project Number 3444-0038

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